

BS – 101 Engineering Physics

Lecture – 17

Engr. Safee
Ullah

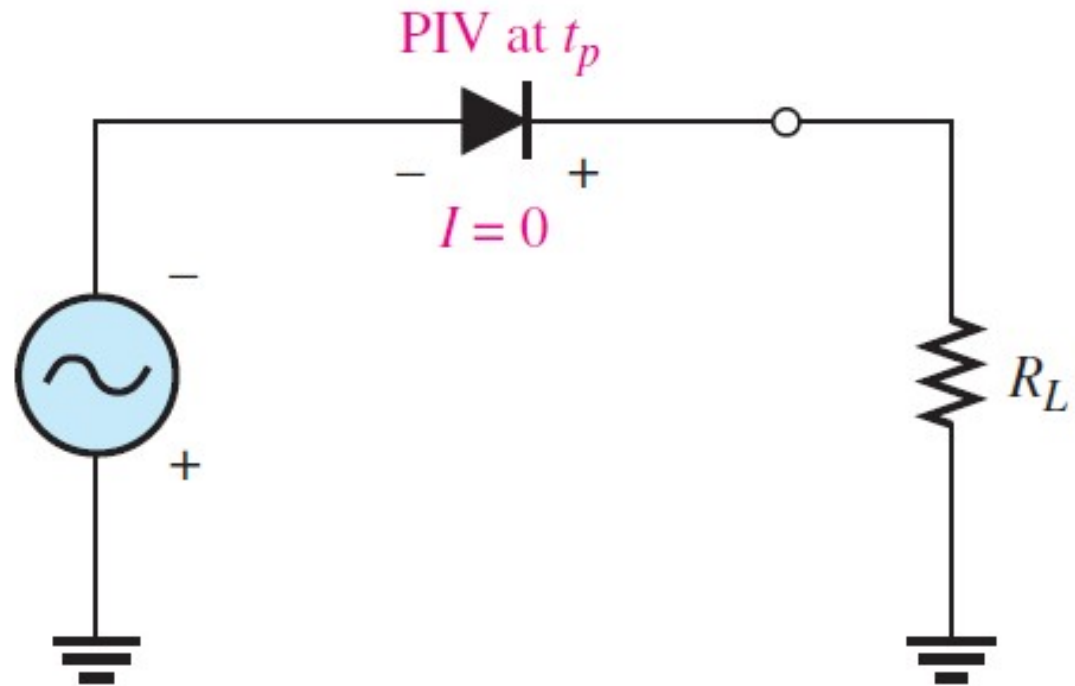
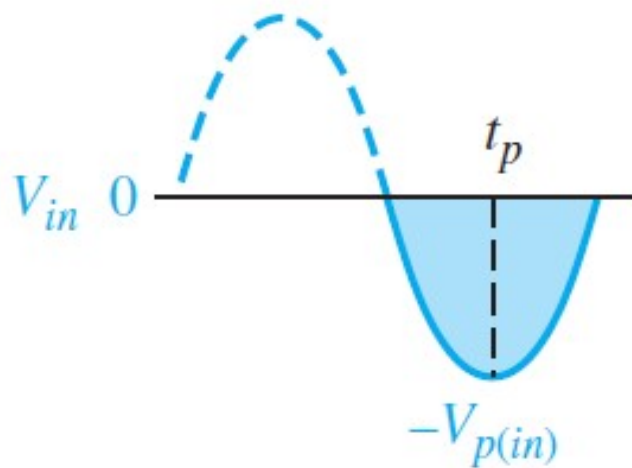


Peak Inverse Voltage (PIV)

- The **peak inverse voltage (PIV)** equals the peak value of the input voltage, and the diode must be capable of withstanding this amount of repetitive reverse voltage.
- For the diode in Figure on next slide, the maximum value of reverse voltage, designated as PIV, occurs at the peak of each negative alternation of the input voltage when the diode is reverse-biased.
- A diode should be rated at least 20% higher than the PIV.

$$PIV = V_{p(in)}$$

Peak Inverse Voltage (PIV) Contd.

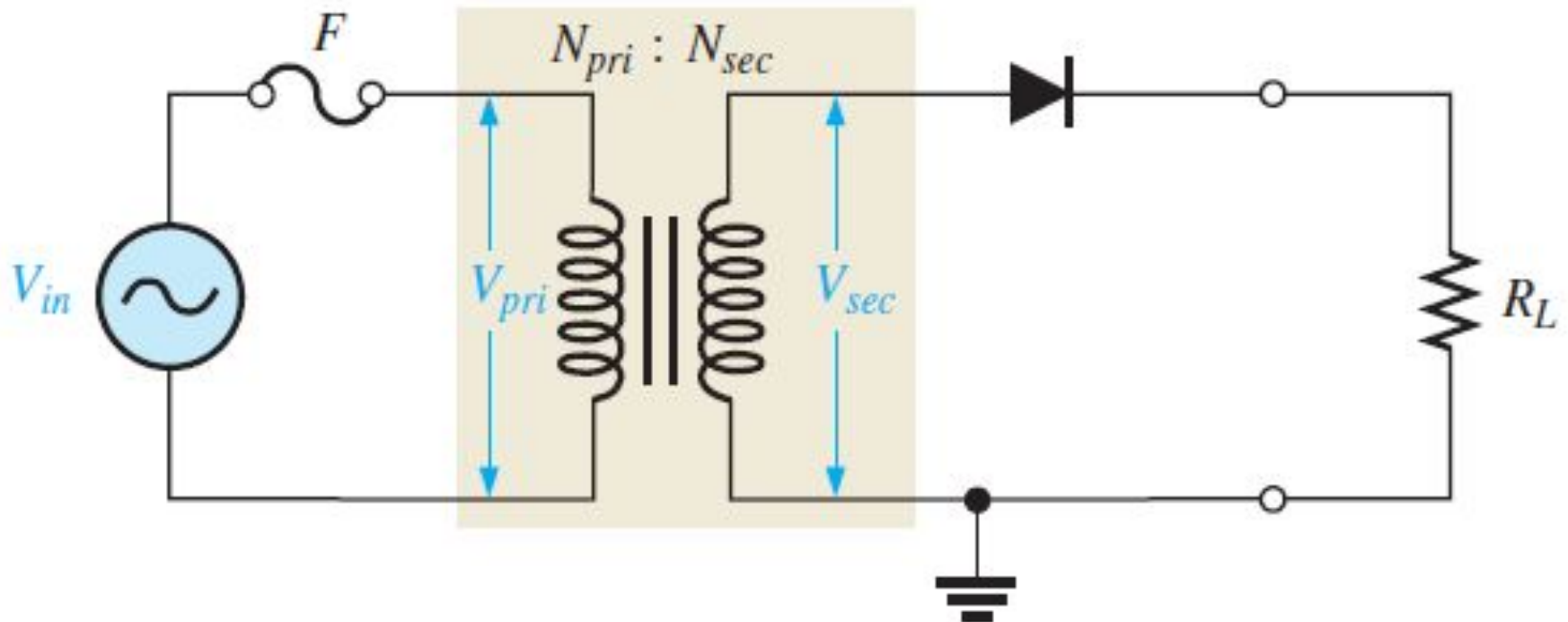




Transformer Coupling

- As you have seen, a transformer is often used to couple the ac input voltage from the source to the rectifier, as shown in Figure on next slide. Transformer coupling provides two advantages.
- First, it allows the source voltage to be stepped down as needed.
- Second, the ac source is electrically isolated from the rectifier, thus preventing a shock hazard in the secondary circuit.

Transformer Coupling (Contd.)



Transformer Coupling (Contd.)

- The amount that the voltage is stepped down is determined by the **turns ratio** of the transformer.
- Unfortunately, the definition of turns ratio for transformers is not consistent between various sources and disciplines.
- In these lectures, we use the definition given by the IEEE for electronic power transformers, which is “the number of turns in the secondary (N_{sec}) divided by the number of turns in the primary (N_{pri}).”
- Thus, a transformer with a turns ratio less than 1 is a step-down type and one with a turns ratio greater than 1 is a step-up type.
- To show the turns ratio on a schematic, it is common practice to show the numerical ratio directly above the windings.
- The secondary voltage of a transformer equals the turns ratio, n , times the primary voltage.

Transformer Coupling (Contd.)

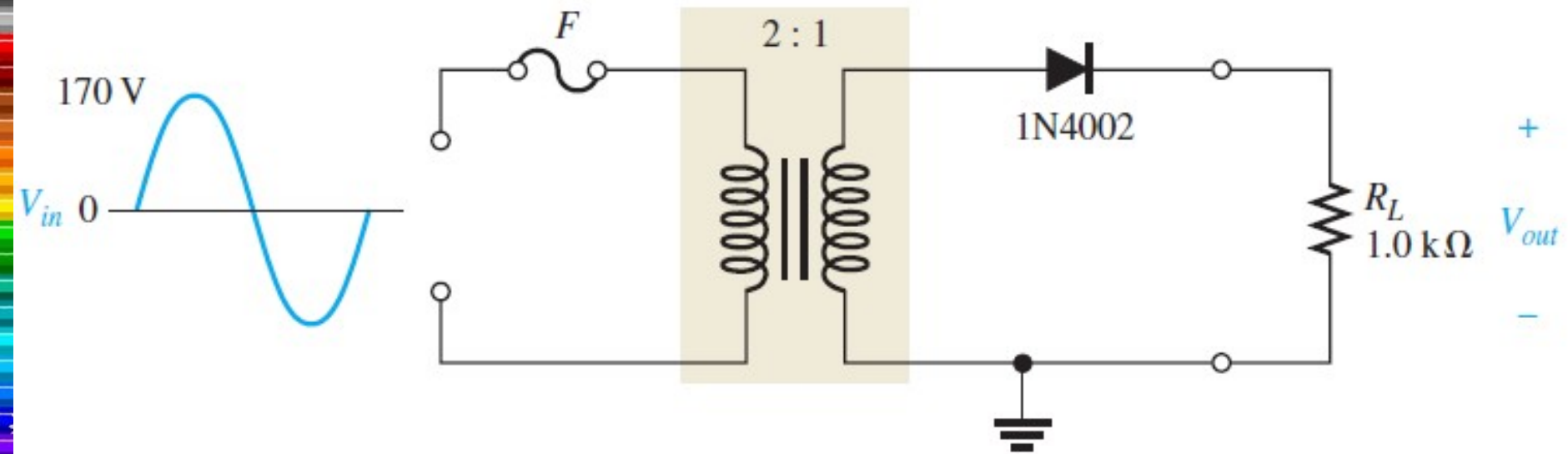
- If $n > 1$, the secondary voltage is greater than the primary voltage.
- If $n < 1$, the secondary voltage is less than the primary voltage.
- If $n = 1$, then $V_{sec} = V_{pri}$.
- The peak secondary voltage, $V_{p(sec)}$, in a transformer-coupled half-wave rectifier is the same as $V_{p(in)}$ in previous equation. Therefore, Equation can be written in terms of $V_{p(sec)}$ is

$$V_{p(out)} = V_{p(sec)} - 0.7 V_{p(sec)}$$
- and PIV in terms of $V_{p(sec)}$ is

$$PIV = V_{p(sec)}$$
- Turns ratio is useful for understanding the voltage transfer from primary to secondary.
- However, transformer datasheets rarely show the turns ratio.
- A transformer is generally specified based on the secondary voltage rather than the turns ratio.

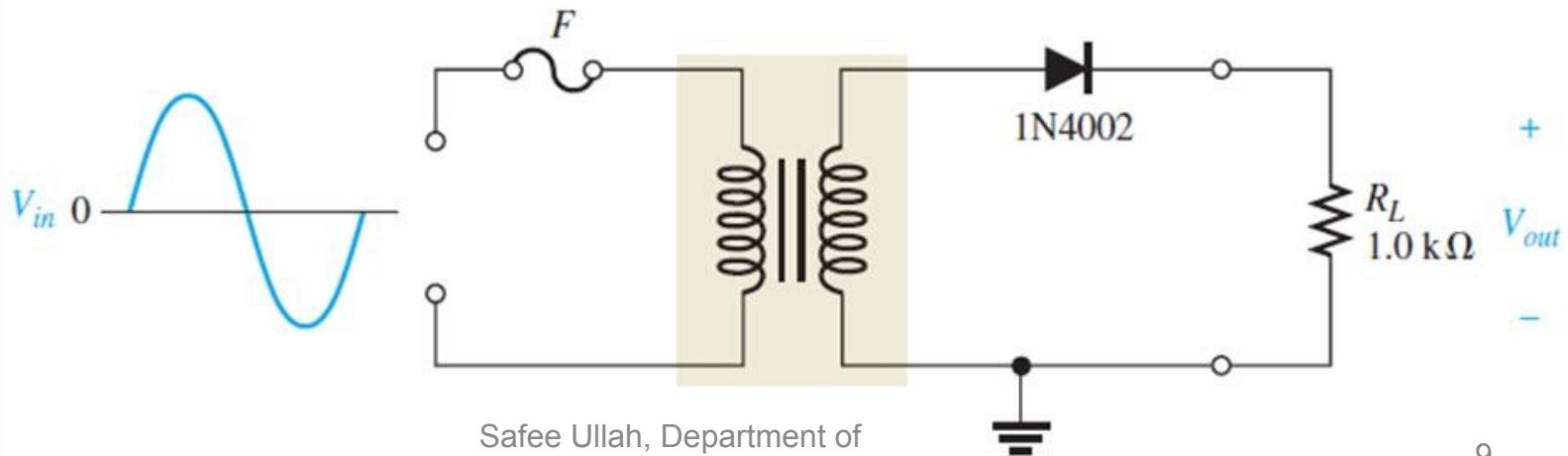
Transformer Coupling (Example)

- Determine the peak value of the output voltage if the turns ratio is 0.5.



Transformer Coupling (Example) Contd.

- Determine the peak value of the output voltage if $n = 2$ and $V_{p(in)} = 312 \text{ V}$.
- What is the PIV across the diode?
- Describe the output voltage if the diode is turned around.

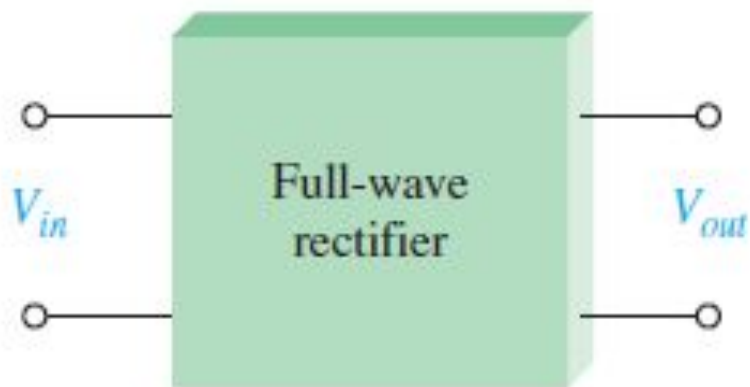
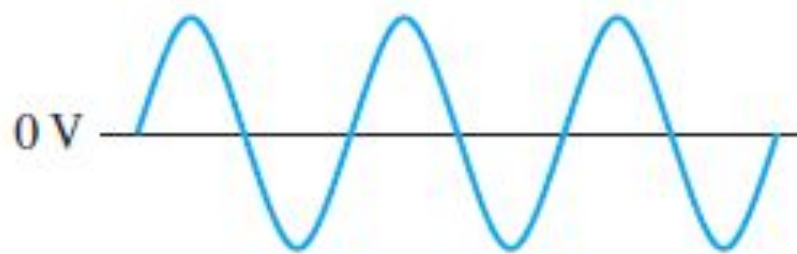




Full Wave Rectification

- A **full-wave rectifier** allows unidirectional (one-way) current through the load during the entire 360° of the input cycle, whereas a half-wave rectifier allows current through the load only during one-half of the cycle. The result of full-wave rectification is an output voltage with a frequency twice the input frequency and that pulsates every half-cycle of the input, as shown in Figure on next slide.

Full Wave Rectification



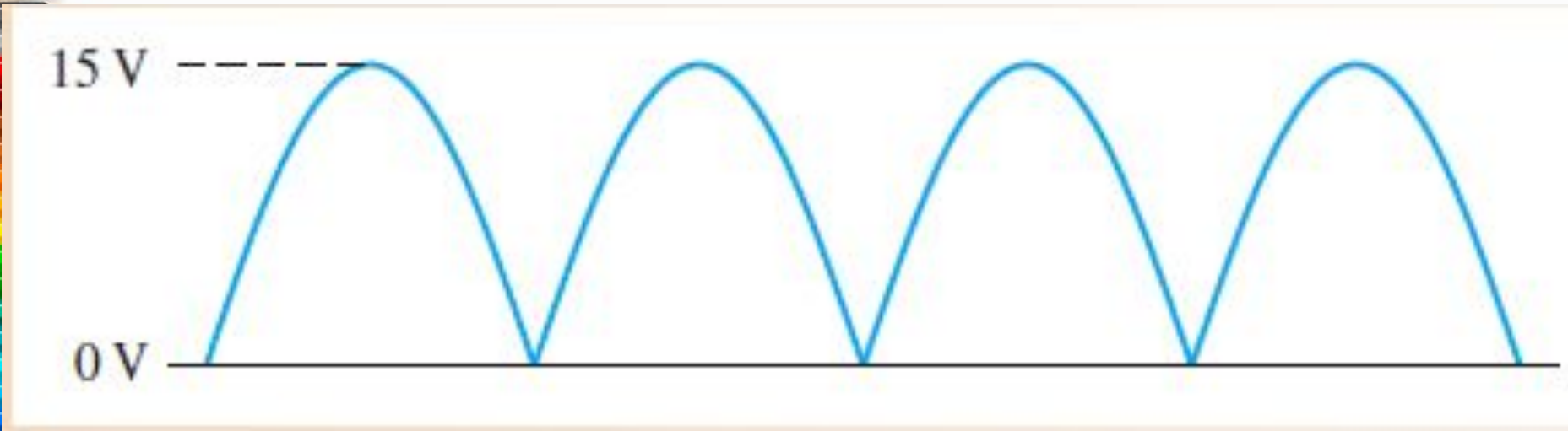
Full Wave Rectification

(Contd.)

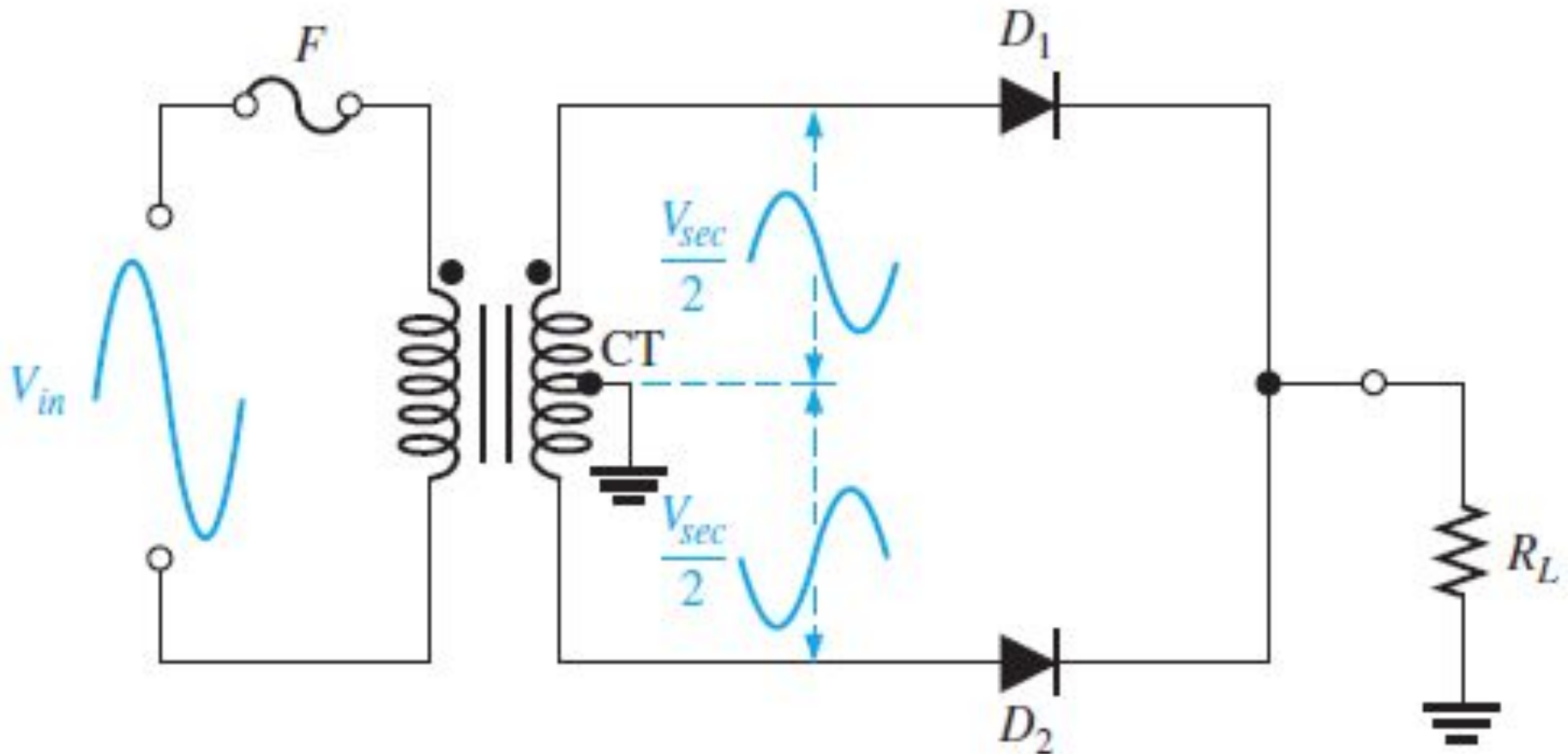
- The number of positive alternations that make up the full-wave rectified voltage is twice that of the half-wave voltage for the same time interval.
- The average value, which is the value measured on a dc voltmeter, for a full-wave rectified sinusoidal voltage is twice that of the half-wave, as shown in the following formula:
$$V_{AVG} = \frac{2V_p}{\pi}$$
- V_{AVG} is approximately 63.7% of V_p for a full-wave rectified voltage.

Full Wave Rectification (Example)

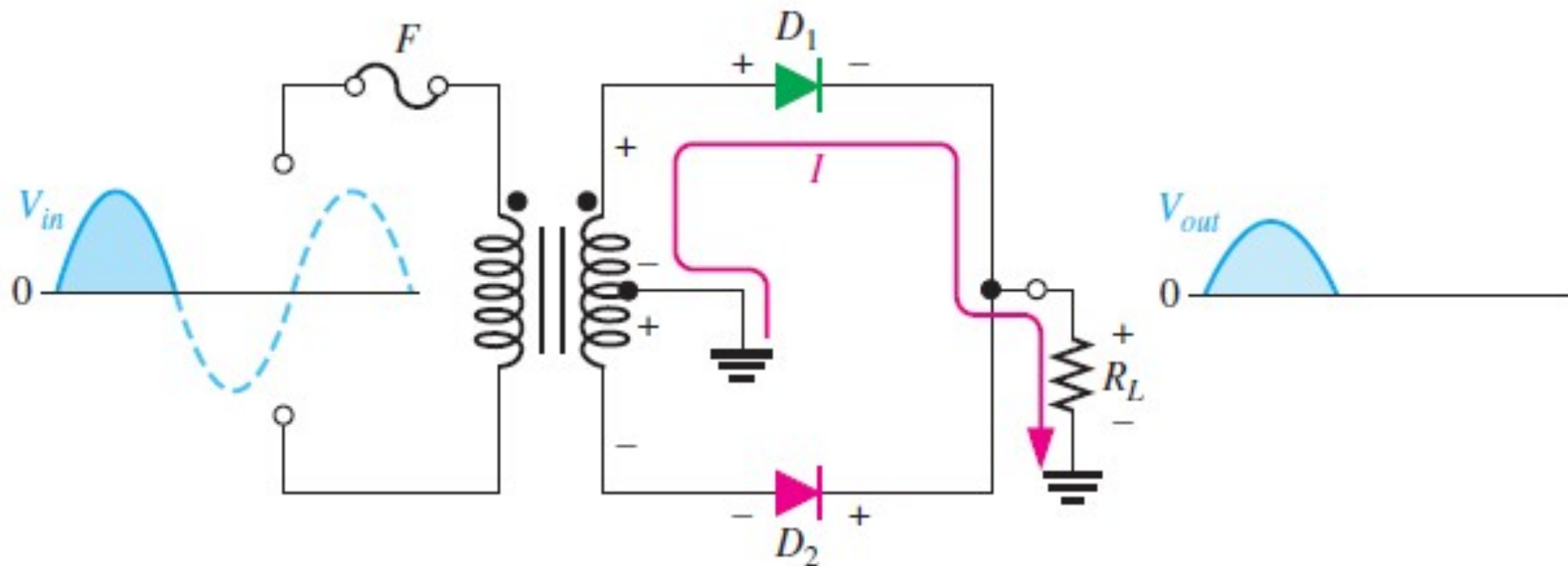
- Find the average value of the full-wave rectified voltage



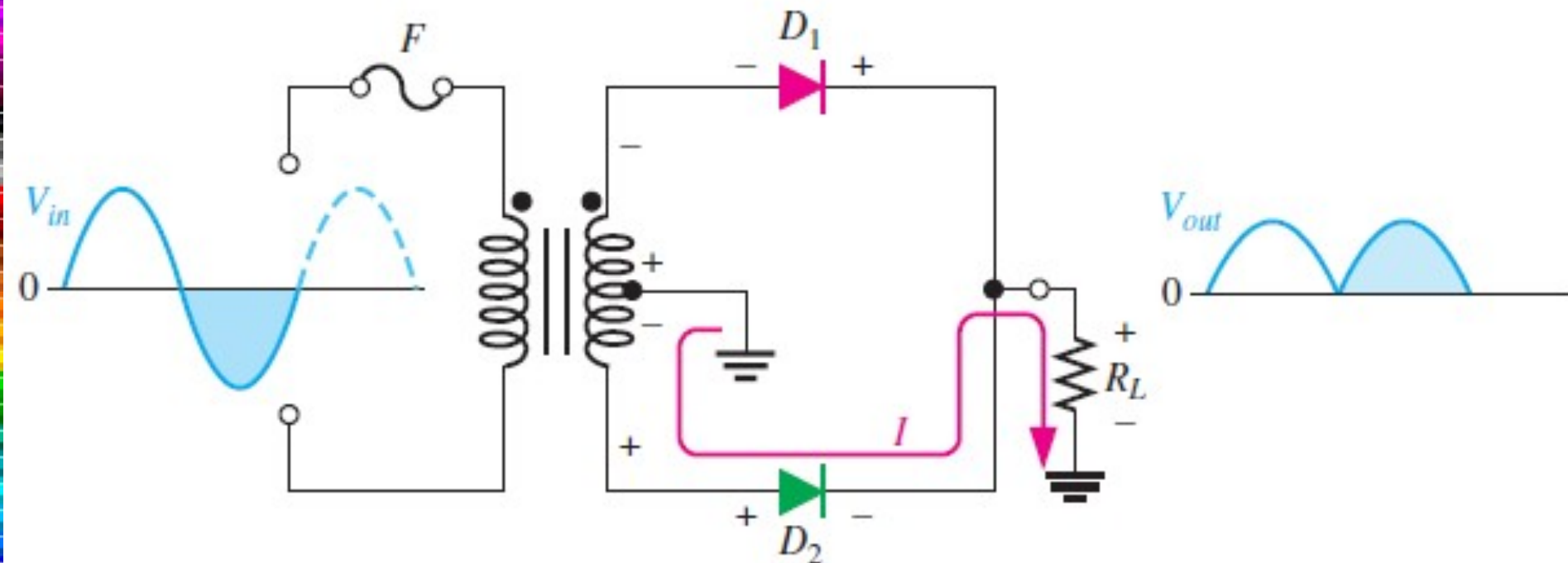
Center-Tapped Full-Wave Rectifier Operation



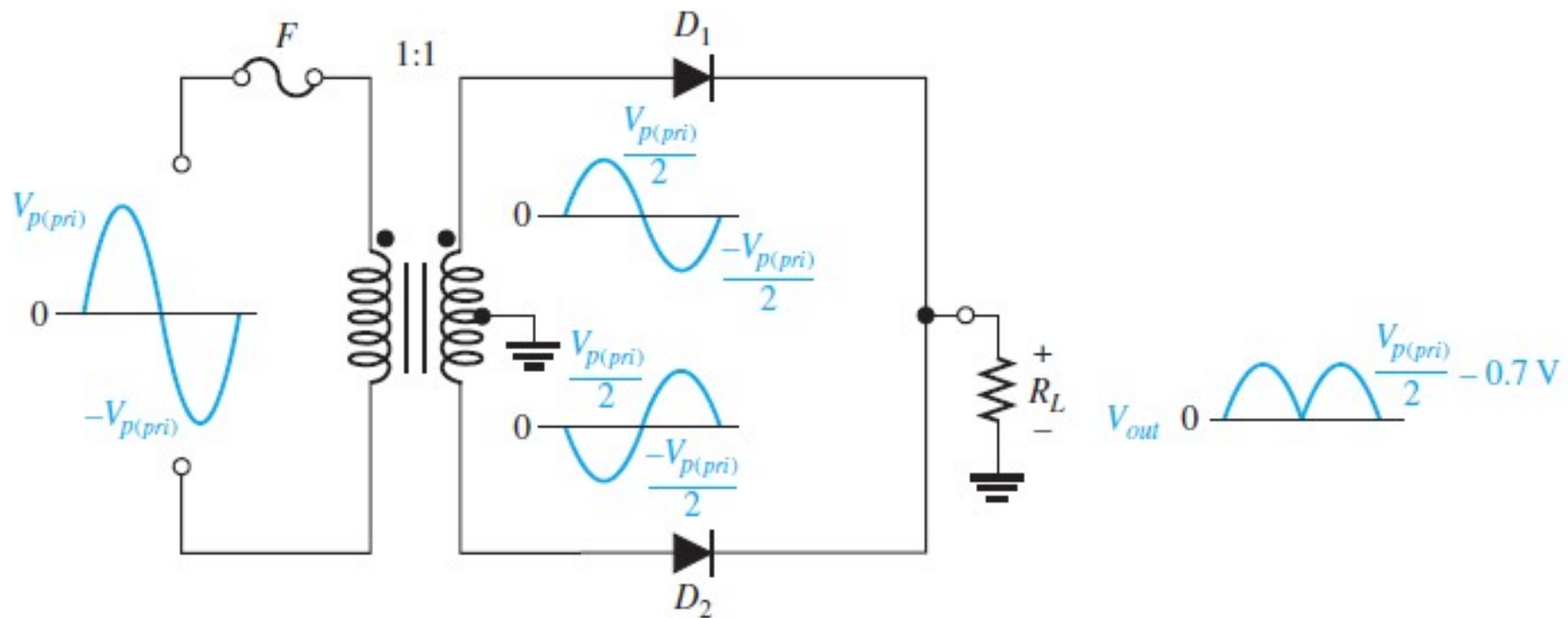
Center-Tapped Full-Wave Rectifier Operation (Contd.)



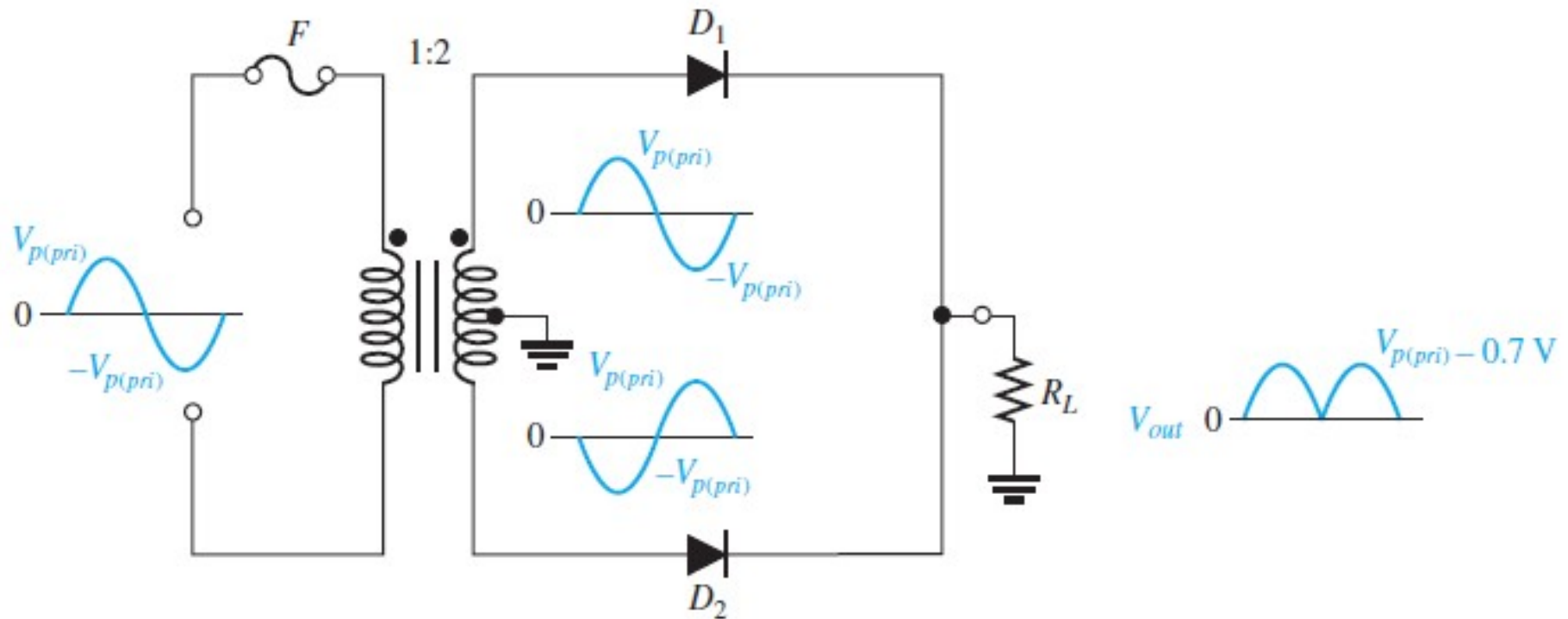
Center-Tapped Full-Wave Rectifier Operation (Contd.)



Effect of the Turns Ratio on the Output Voltage

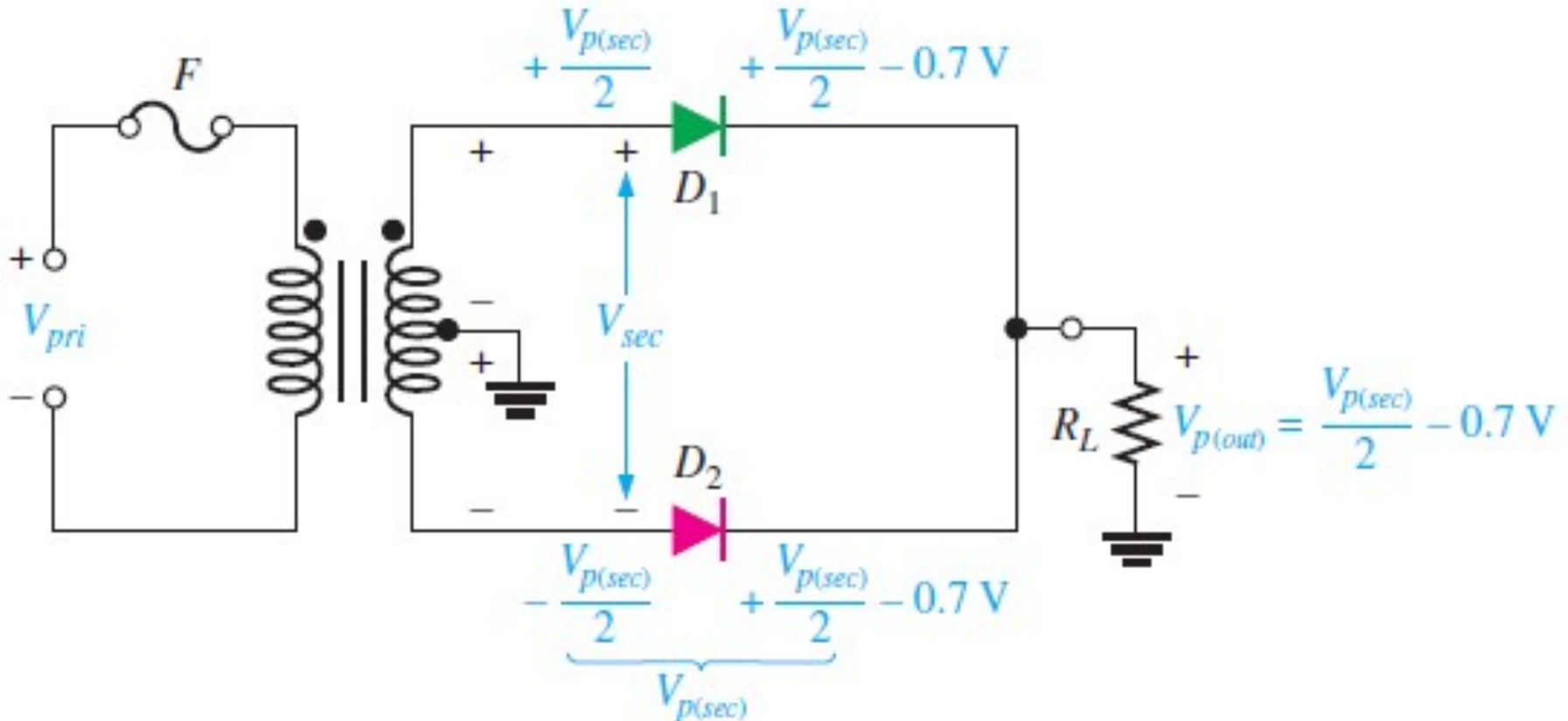


Effect of the Turns Ratio on the Output Voltage (Contd.)



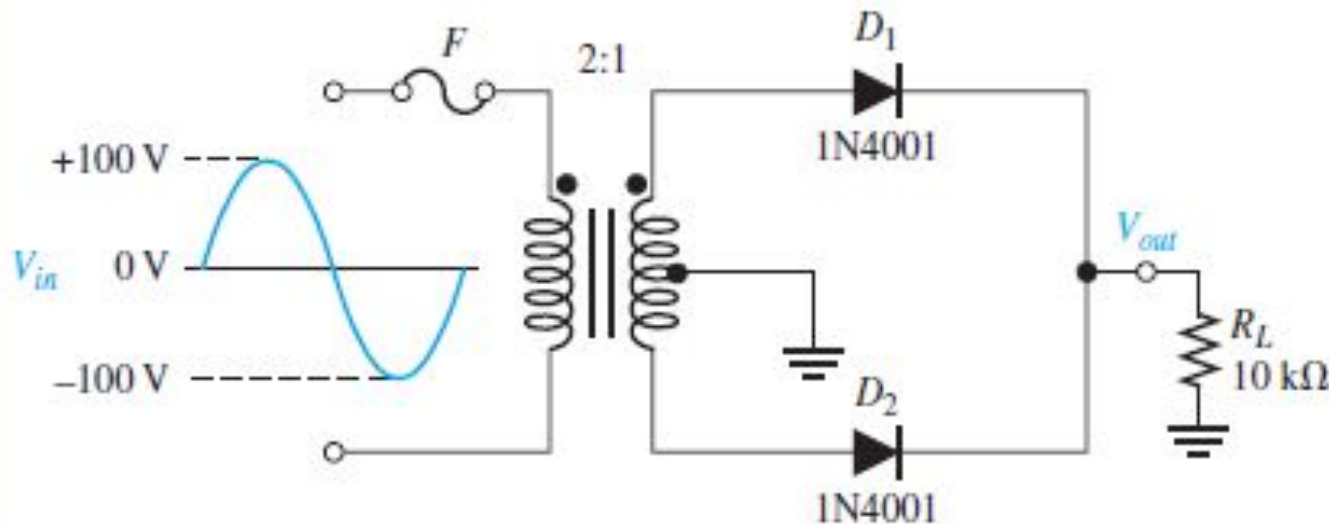
Peak Inverse Voltage

- $PIV = 2V_{p(out)} + 0.7 V$



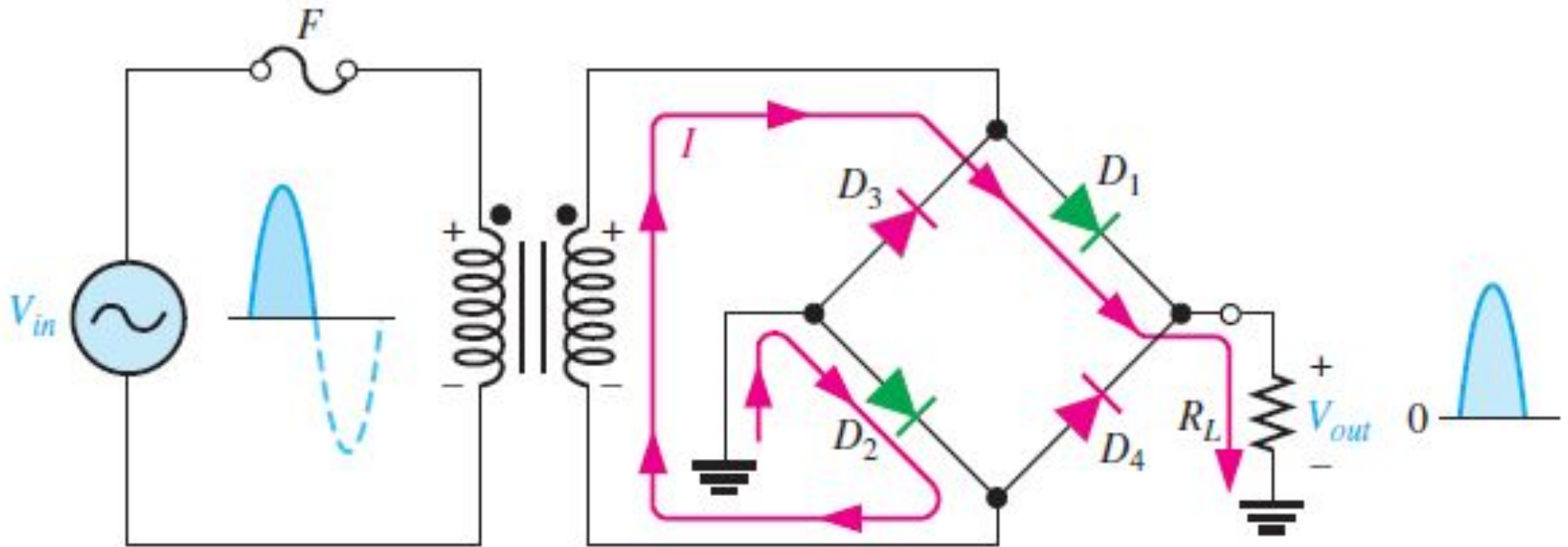
Example

- Show the voltage waveforms across each half of the secondary winding and across R_L when a 100 V peak sine wave is applied to the primary winding.

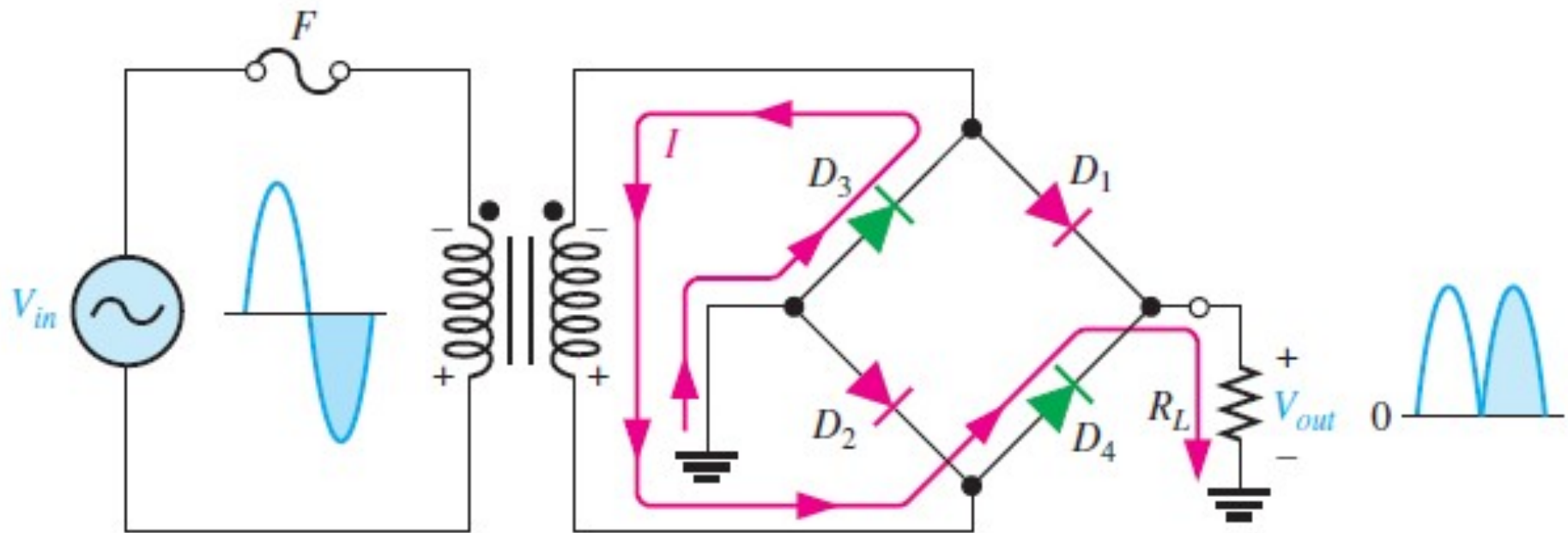


Bridge Full Wave Rectifier Operation

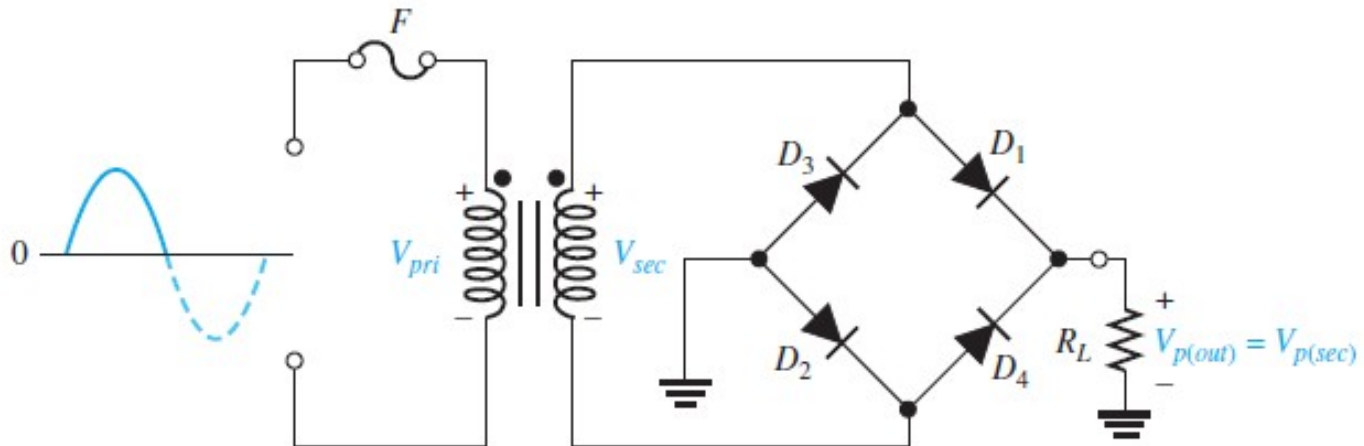
- The **bridge rectifier** uses four diodes connected as shown in Figure



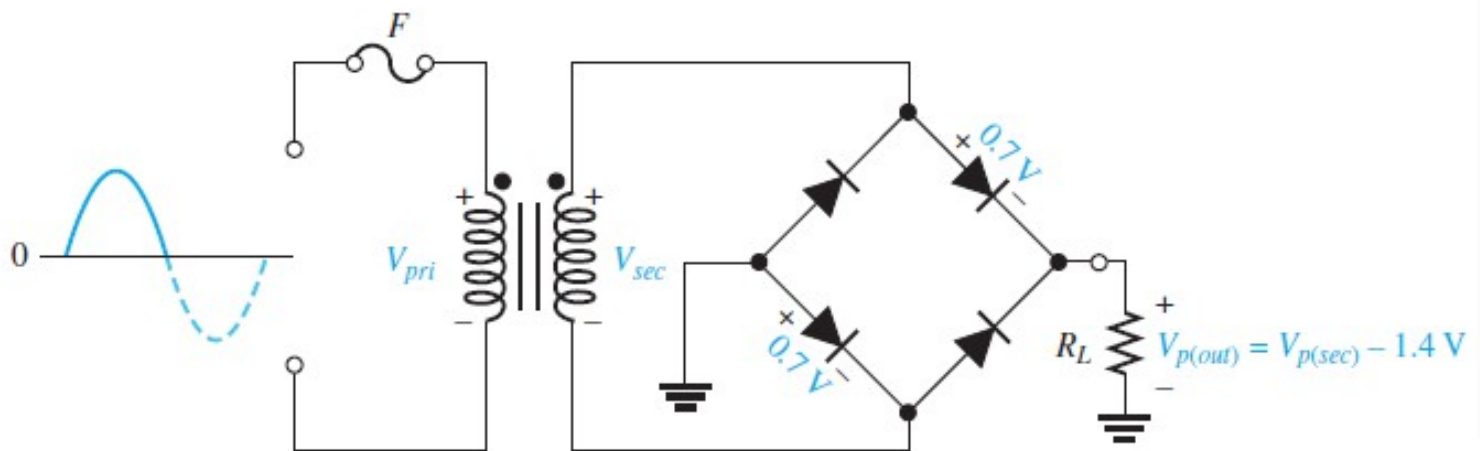
Bridge Full Wave Rectifier Operation (Contd.)



Output



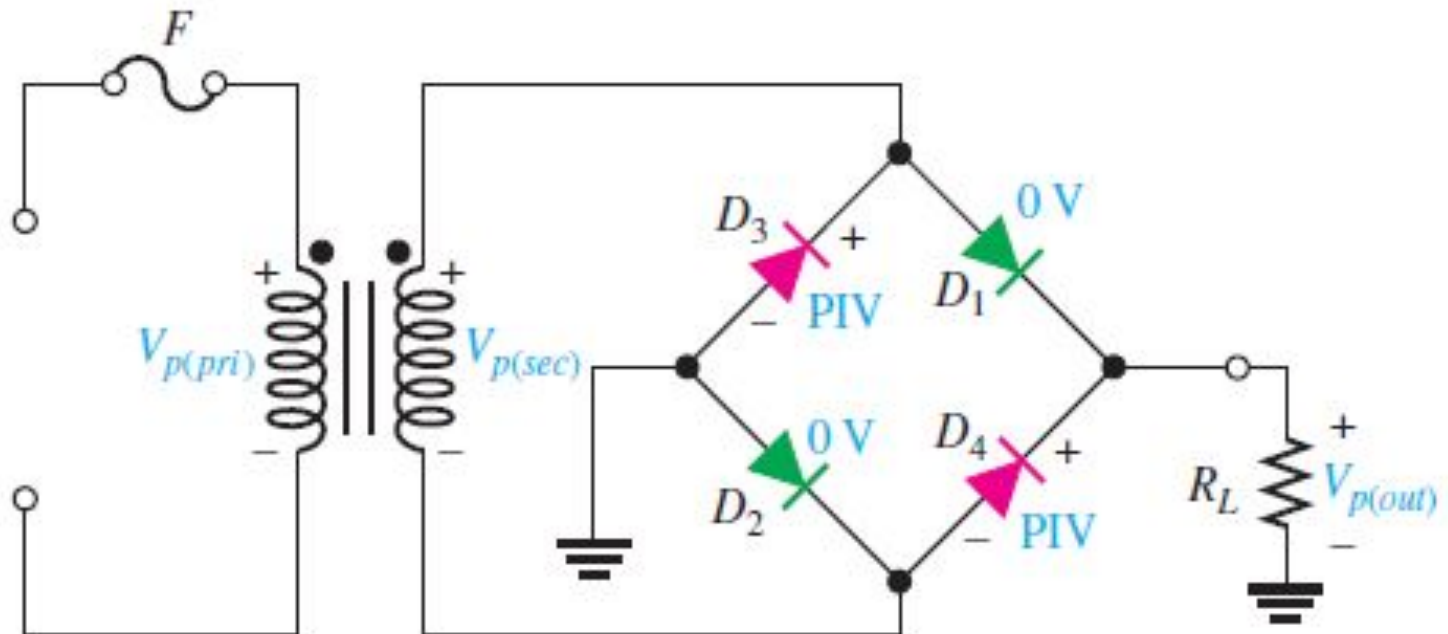
(a) Ideal diodes



(b) Practical diodes (Diode drops included)

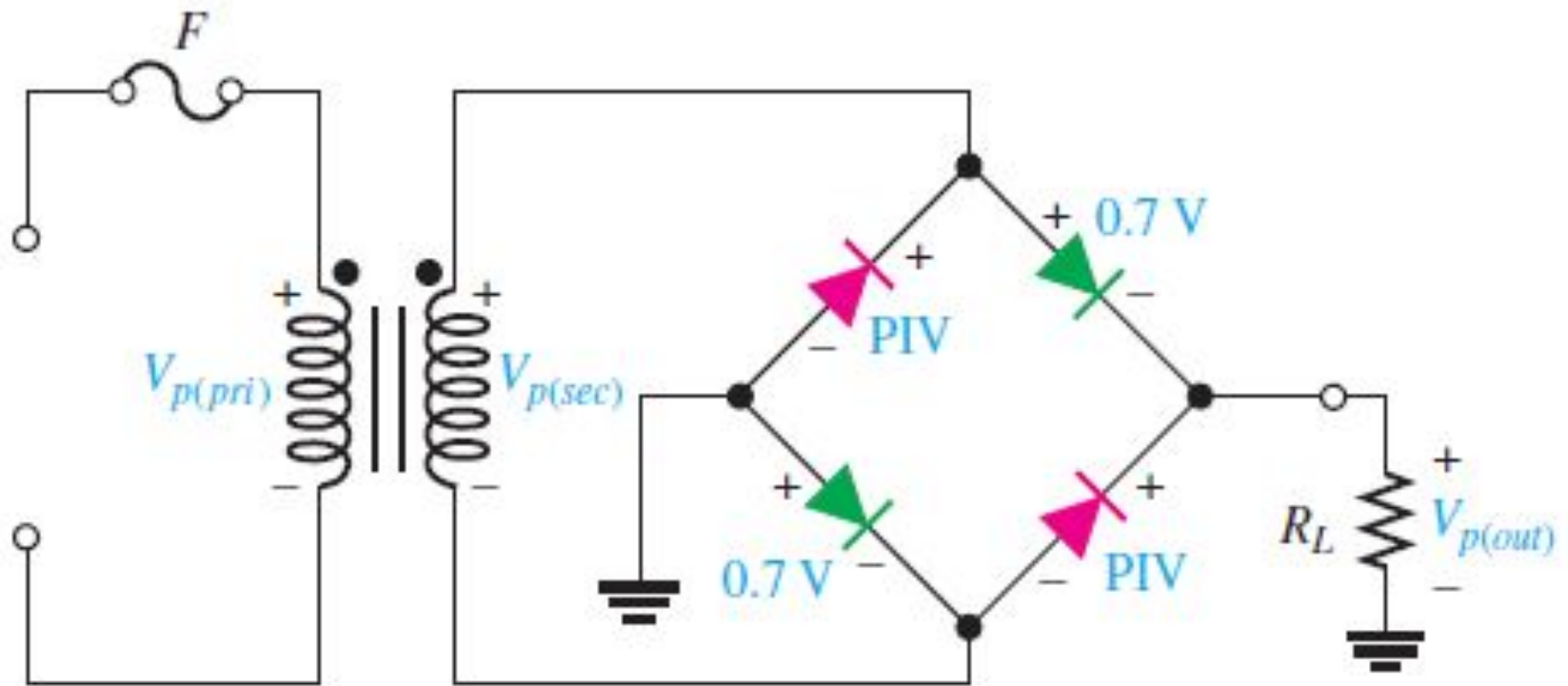
Peak Inverse Voltage (Ideal Diode)

$$PIV = V_{p(out)}$$



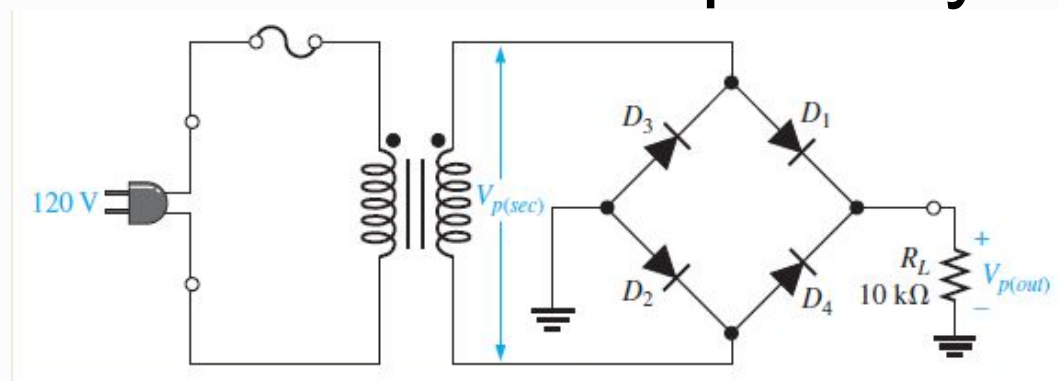
Peak Inverse Voltage (Practical Diode)

$$PIV = V_{p(out)} + 0.7 V$$



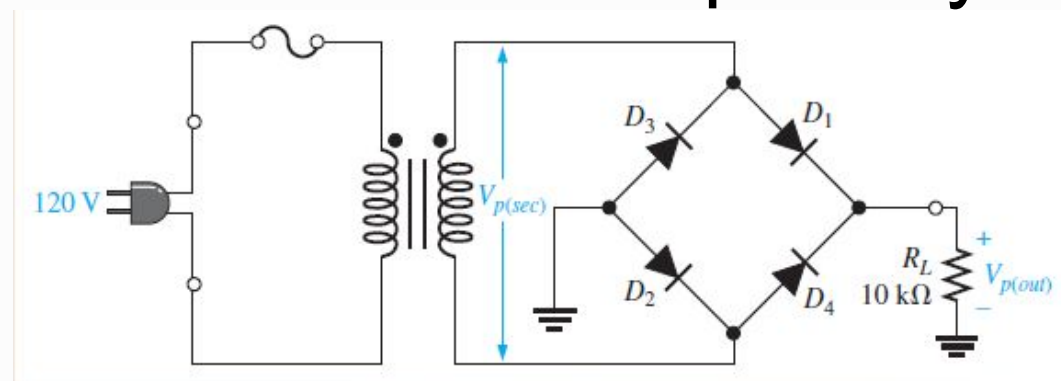
Example

- Determine the peak output voltage for the bridge rectifier. Assuming the practical model, what PIV rating is required for the diodes? The transformer is specified to have a 12 V_{rms} secondary voltage for the standard 120 V across the primary.



Example (Contd.)

- Determine the peak output voltage for the bridge rectifier. Assuming the practical model, what PIV rating is required for the diodes? The transformer is specified to have a 30 V_{rms} secondary voltage for the standard 120 V across the primary.

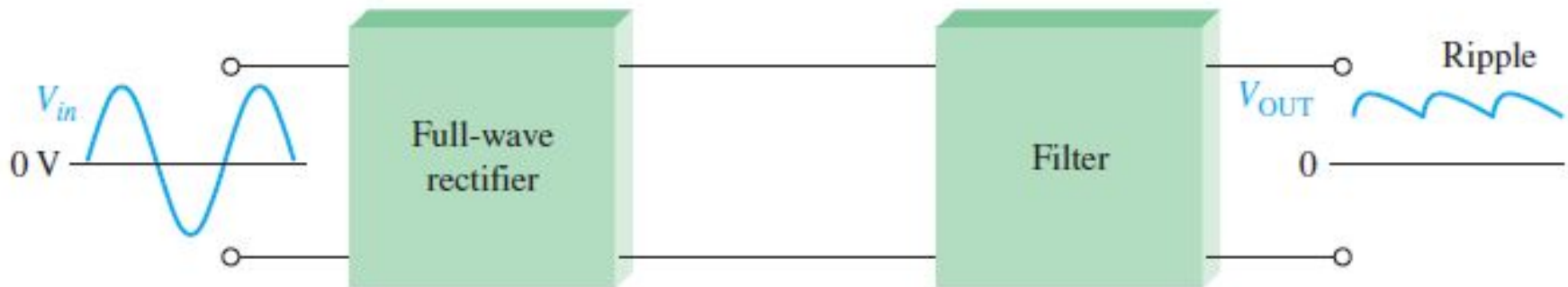




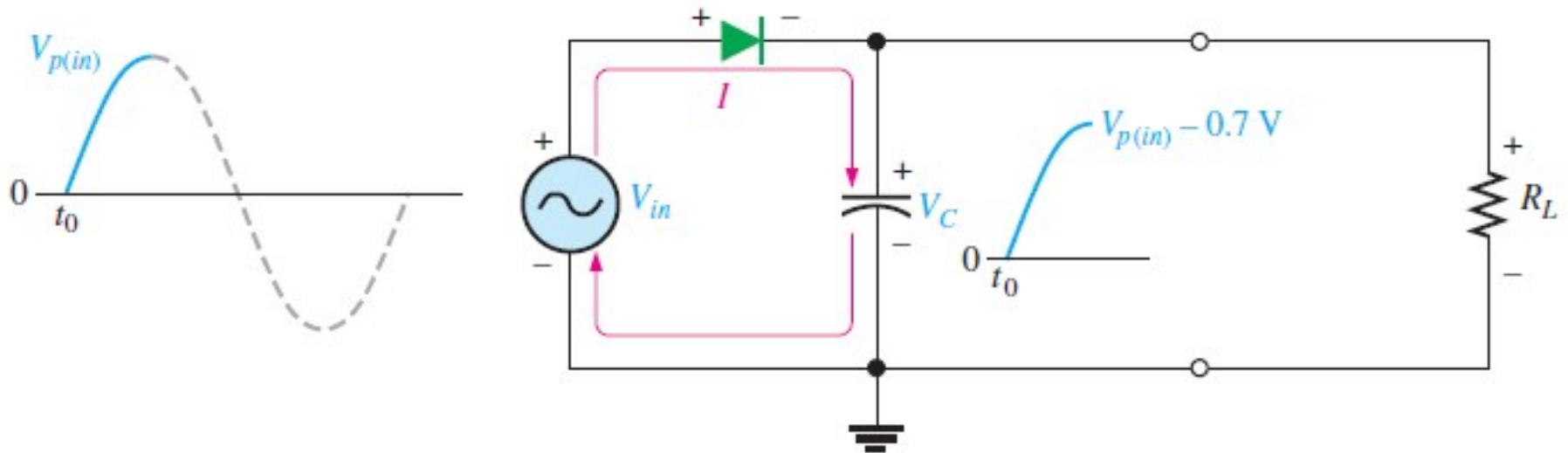
Power Supply Filter

- In most power supply applications, the standard 50 Hz ac power line voltage must be converted to an approximately constant dc voltage.
- The 50 Hz pulsating dc output of a half-wave rectifier must be filtered to reduce the large voltage variations.
- Figure on next slide illustrates the filtering concept showing a nearly smooth dc output voltage from the filter.
- The small amount of fluctuation in the filter output voltage is called *ripple*.

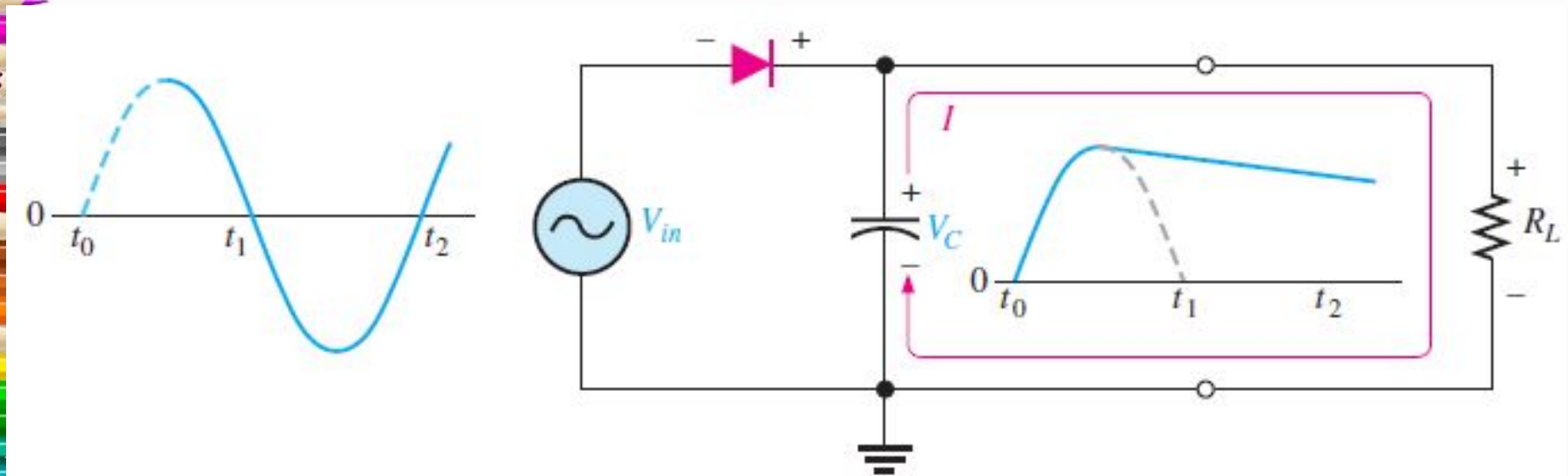
Power Supply Filter (Contd.)



Power Supply Filter (Contd.)



Power Supply Filter (Contd.)

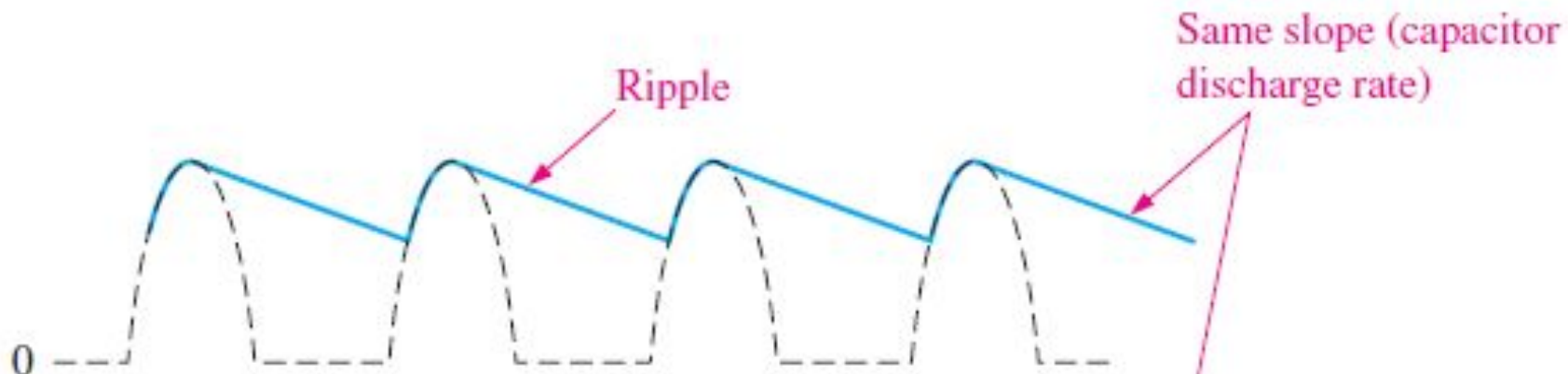




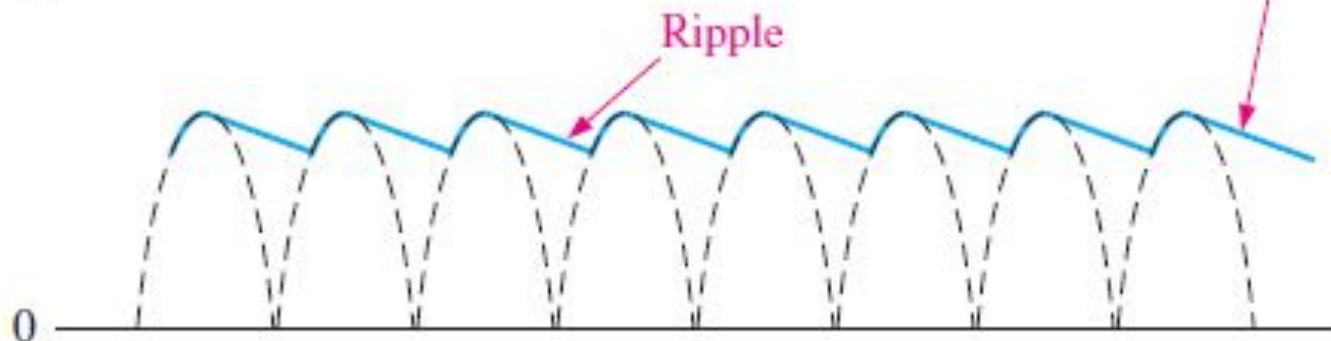
Ripple Voltage

- As you have seen, the capacitor quickly charges at the beginning of a cycle and slowly discharges through R_L after the positive peak of the input voltage (when the diode is reverse-biased).
- The variation in the capacitor voltage due to the charging and discharging is called the **ripple voltage**.
- Generally, ripple is undesirable; thus, the smaller the ripple, the better the filtering action.

Ripple Voltage (Contd.)



(a) Half-wave



(b) Full-wave



Ripple Factor

- The **ripple factor (r)** is an indication of the effectiveness of the filter and is defined as

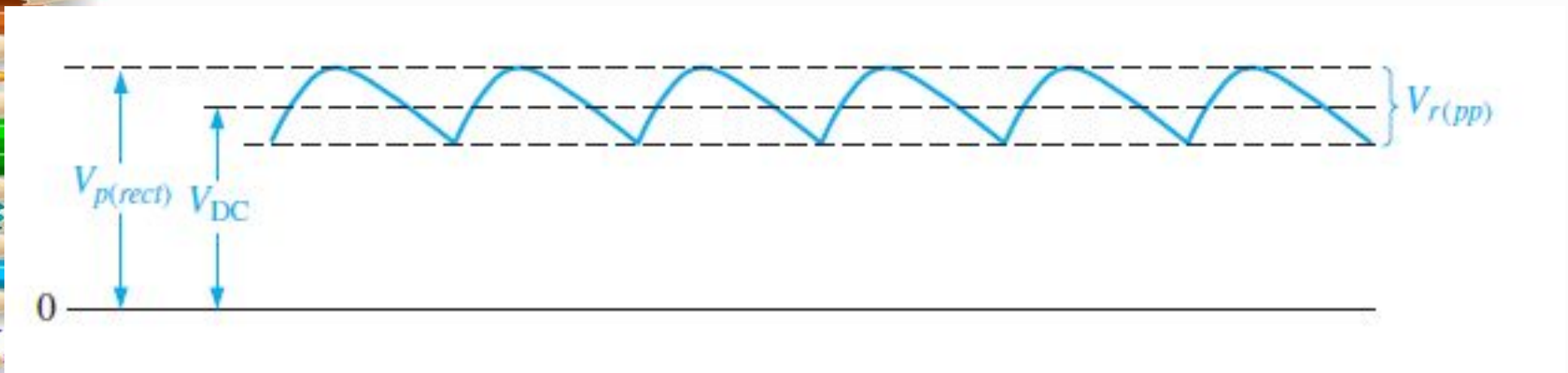
$$r = \frac{V_{r(pp)}}{V_{DC}}$$

- Where
- $V_{r(pp)}$ = peak to peak ripple voltage
- V_{DC} = dc (average) value of the filter's output voltage.

Ripple Factor (Contd.)

- The **ripple factor (r)** is an indication of the effectiveness of the filter and is defined as

$$r = \frac{V_{r(pp)}}{V_D}$$





Part of Assignment - 3

$$V_{r(pp)} \cong \left(\frac{1}{fR_L C} \right) V_{p(rect)}$$

$$V_{DC} \cong \left(1 - \frac{1}{2fR_L C} \right) V_{p(rect)}$$

The derivations for these equations can be found in “Derivations of Selected Equations” at www.pearsonhighered.com/floyd.

Example

- Determine the ripple factor for the filtered bridge rectifier with a load.

